

# MIDTERM EXAM

## ECON301—ECONOMETRICS I

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ALEJANDRO MELO PONCE

NAZARBAYEV UNIVERSITY

1. Let  $X$  have a uniform $[0,1]$  distribution and  $Z$  have a uniform $[0, \frac{1}{100}]$  distribution. Suppose  $X$  and  $Z$  are independent. Let  $Y = X + Z$ . Consider the random vector  $(X, Y)$ .
  - (a) *10 points.* Derive the joint PDF of  $(X, Y)$ .
  - (b) *10 points.* Derive the conditional PDF of  $Y$  given  $X$ .
  - (c) *10 points.* Compute the conditional expectation function (CEF)  $E(Y|X)$ .
  - (d) *10 points.* Compute the best linear predictor (BLP)  $L(Y|X)$ .
2. Consider the random vector  $(X, Y)$  describing a population of interest. Suppose that  $X$  is Bernoulli random variable that equals one with probability  $p$  and is zero otherwise. Furthermore, suppose that conditional on the value of  $X$ , the random variable  $Y$  is distributed according to:

$$Y|X = 0 \sim \text{Normal}(\mu_0, \sigma_0^2), \quad Y|X = 1 \sim \text{Normal}(\mu_1, \sigma_1^2).$$

The unknown parameters  $\mu_0$  and  $\mu_1$  are not necessarily different to each other. Similarly, the unknown parameters  $\sigma_0^2$  and  $\sigma_1^2$  are not necessarily different to each other.

- (a) *15 points.* Let  $L(Y|X)$  be the best linear predictor. Show that the  $L(Y|X)$  is equal to the CEF of  $Y$  given  $X$  by showing that  $\beta_0 = E(Y|X = 0)$  and  $\beta_1 = E(Y|X = 1) - E(Y|X = 0)$ .
- (b) *15 points.* Suppose that you have a random sample  $(X_i, Y_i)$ ,  $i = 1, \dots, n$ , from the population described above. Discuss the extent to which assumptions A1–A5 from topic 3 are satisfied.
- (c) *15 points.* Suppose that  $\sigma_0^2 = \sigma_1^2$  and denote this common value by  $\sigma^2$ . Propose a method to estimate and do statistical inference about the parameters  $\mu_0$ ,  $\mu_1$ , and  $\sigma^2$ .

3. Shynar has data on 7,000 students in the United States who were in eight grade in 1988. Her data set includes the variables *math12*, and *read12*, which are scores on twelfth grade standardized math and reading tests, respectively. Shynar would like to estimate a simple linear regression explaining *math12* on *read12*. The results from her estimation as well as some sample statistics are presented below.

|               | Estimate | Std. Error |
|---------------|----------|------------|
| Intercept     | 15.15304 | 0.43204    |
| <i>read12</i> | 0.71429  | 0.00821    |

$$n = 7430, \quad \hat{\sigma} = 6.65763, \quad \overline{math12} = 52.13362, \quad \overline{read12} = 51.7724,$$

$$\sum_{i=1}^{7430} read12_i^2 = 20572747, \quad \sum_{i=1}^{7430} math12_i^2 = 20858815,$$

$$\sum_{i=1}^{7430} read12_i \times math12_i = 20523840.$$

Suppose that *all* assumptions 1 to 5 from the conditional approach presented in lecture 3 hold.

- 5 points. What is the estimated ceteris paribus effect of *read12* on *math12*? How do you interpret it?
- 5 points. Does the intercept reported in the results above have a meaningful interpretation? Explain your answer.
- 10 points. Suppose that Shynar wants to test the following hypotheses:  $H_0$  : “The effect of 10 more points in the reading test is *exactly* an increase of 7.1 points in the math test” versus  $H_a$  : “The effect of 10 more points in the reading test is an increase of *more than* 7.1 points in the math test. ” Construct the previous hypothesis test and test it at the 5% and 1% significance levels.
- 10 points. Construct a 99% confidence interval for the ceteris paribus effect *read12* on *math12*.
- 15 points. Suppose that Shynar presents her findings to a superintendent of a school district, and the superintendent says, “Your findings show that to improve math scores we just need to improve reading scores, so we should hire more reading tutors.” How would you respond to this comment? In order to answer this question properly, you will need to compute the regression of *read12* on *math12*. This regression can be computed from the data I gave you above.

# Useful formulas & Tail probabilities

## Uniform Distribution

Let  $W$  be a random variable that is  $\text{Uniform}[a, b]$ .

- PDF:  $f(w) = \frac{1}{b-a} I_{[a,b]}(w)$ .
- CDF:  $F(w) = \begin{cases} 0 & w < a \\ \frac{w-a}{b-a} & w \in [a, b] \\ 1 & w > b \end{cases}$
- Mean:  $\frac{a+b}{2}$ .
- Variance:  $\frac{(b-a)^2}{12}$ .

## Bernoulli Distribution

Let  $D$  be a random variable that is  $\text{Bernoulli}(p)$ .

- PMF:  $D = 1$  with probability  $p$ ,  $D = 0$  with probability  $1 - p$ .
- Mean:  $p$ .
- Variance:  $p(1 - p)$ .

## Normal Distribution

Let  $Z$  be random variable that is  $\text{Normal}(\mu, \sigma^2)$ .

- PDF:  $f(z) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{z-\mu}{\sigma}\right)^2}$ .
- Mean:  $\mu$ .
- Variance:  $\sigma^2$ .

## Tail probabilities and cutoff values for the $t$ -student Distribution

Let  $T$  be a random variable with a  $t$ -student distribution with 7428 degrees of freedom.

| $\tau$ | $\Pr(T > \tau)$ |
|--------|-----------------|
| 1.28   | 0.1             |
| 1.65   | 0.05            |
| 1.96   | 0.025           |
| 2.33   | 0.01            |
| 2.58   | 0.005           |

Table 1: Right-tail probabilities and cutoff values for  $t_{(7428)}$ .